

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)

19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. **Propose a frequency allocation strategy for mixed wireless services.** (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I Nithish kumar D(192512329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Nithish kumar d

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

Frequency Allocation Strategy

- **Frequency allocation** means dividing available radio spectrum among different wireless services.
- Different services like **IoT, 4G, 5G, Wi-Fi and Satellite** need different frequency bands.
- **Separate frequency bands** are assigned to each service to reduce interference.
- **Guard bands** are kept between adjacent services to prevent unwanted signal overlap.
- **Priority-based allocation** can be used for important services such as emergency communication.
- **Dynamic spectrum allocation** allows unused frequency bands to be temporarily given to high-traffic services.
- **Interference monitoring** helps detect and reduce overlapping signals.
- The strategy improves **spectrum utilization** and reduces wastage.
- It helps maintain better **Quality of Service (QoS)** for different wireless applications.
- **MATLAB simulation** can be used to visualize frequency bands and calculate spectrum utilization.

Simple Workflow

Available Spectrum → Divide into Bands → Assign Services → Add Guard Bands → Monitor Traffic → Dynamic Reallocation

MATLAB CODE:

```
clc;  
clear;  
close all;
```

%% Frequency Allocation Strategy for Mixed Wireless Services

% Frequency range

```
f_start = 700;    % MHz
```

```
f_end   = 6000;   % MHz
```

%% Define wireless services

```
services = {  
    'IoT';  
    'LTE/4G';  
    '5G';  
    'Wi-Fi';  
    'Satellite'
```

```

};

% Allocated frequency bands (MHz)
f_low = [700 1800 3500 5000 5800];
f_high = [900 2200 3800 5200 6000];

% Bandwidth
BW = f_high - f_low;

%% Display allocation table
fprintf('Frequency Allocation Strategy\n');
fprintf('-----\n');

for i = 1:length(services)
    fprintf('%-12s : %4d - %4d MHz | BW = %4d MHz\n', ...
        services{i}, f_low(i), f_high(i), BW(i));
end

%% Plot frequency allocation
figure('Name','Mixed Wireless Services Frequency Allocation');

for i = 1:length(services)

    rectangle('Position',[f_low(i), i-0.35, BW(i), 0.7], ...
        'FaceColor',[0.3 0.6 0.8], ...
        'EdgeColor','k');

    text((f_low(i)+f_high(i))/2, i, services{i}, ...
        'HorizontalAlignment','center', ...
        'FontWeight','bold');

end

xlabel('Frequency (MHz)');
ylabel('Wireless Services');
title('Frequency Allocation for Mixed Wireless Services');
xlim([f_start f_end]);

```

```

ylim([0 length(services)+1]);
grid on;
yticks(1:length(services));
yticklabels(services);

%% Calculate spectrum utilization
totalSpectrum = f_end - f_start;
allocatedSpectrum = sum(BW);

utilization = (allocatedSpectrum / totalSpectrum) * 100;

fprintf('\nTotal Available Spectrum = %d MHz\n',totalSpectrum);
fprintf('Allocated Spectrum      = %d MHz\n',allocatedSpectrum);
fprintf('Spectrum Utilization      = %.2f %%\n',utilization);

%% Interference check
fprintf('\nInterference Analysis\n');
fprintf('-----\n');

for i = 1:length(services)-1

    if f_high(i) <= f_low(i+1)
        fprintf('%s -> %s : No Overlap\n', ...
            services{i},services{i+1});
    else
        fprintf('%s -> %s : Interference!\n', ...
            services{i},services{i+1});
    end

end

end

%% Dynamic allocation example
% Increase 5G bandwidth during high traffic

traffic_5G = 85;    % Percentage traffic load

if traffic_5G > 70

```

```

fprintf('\nHigh 5G traffic detected.\n');
fprintf('Dynamic allocation activated.\n');

extraBW = 200;    % MHz

f_high(3) = f_high(3) + extraBW;
BW(3) = f_high(3) - f_low(3);

fprintf('5G bandwidth increased to %d MHz.\n',BW(3));
else
    fprintf('\nNormal traffic condition.\n');
end

%% Spectrum utilization after dynamic allocation
allocatedSpectrum2 = sum(BW);
utilization2 = (allocatedSpectrum2 / totalSpectrum) * 100;

fprintf('New Spectrum Utilization = %.2f %%\n',utilization2);

%% Final allocation plot
figure('Name','Dynamic Frequency Allocation');

for i = 1:length(services)

    rectangle('Position',[f_low(i), i-0.35, BW(i), 0.7], ...
        'FaceColor',[0.5 0.7 0.5], ...
        'EdgeColor','k');

    text((f_low(i)+f_high(i))/2, i, services{i}, ...
        'HorizontalAlignment','center', ...
        'FontWeight','bold');

end

xlabel('Frequency (MHz)');
ylabel('Services');
title('Dynamic Frequency Allocation Strategy');

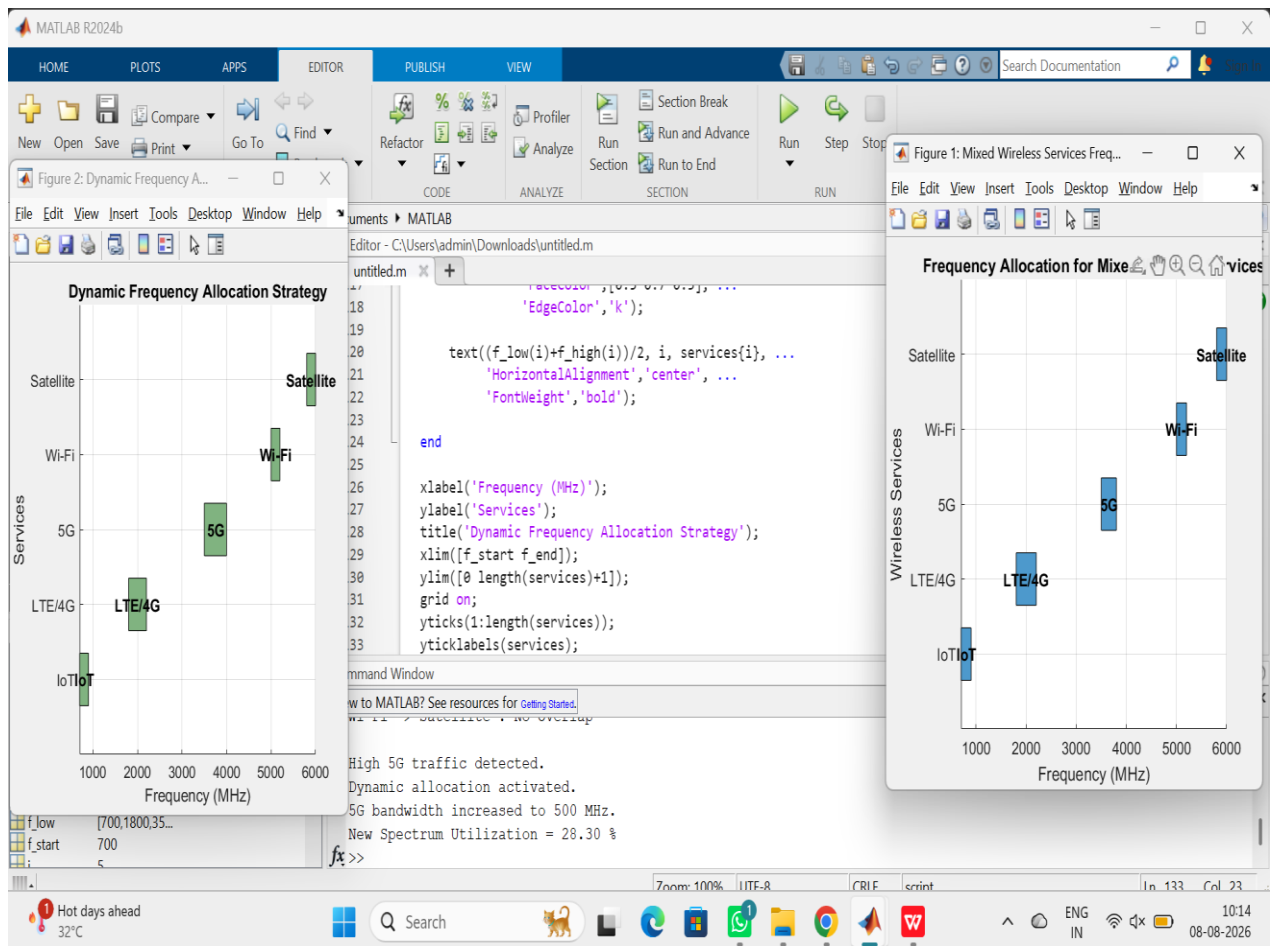
```

```

xlim([f_start f_end]);
ylim([0 length(services)+1]);
grid on;
yticks(1:length(services));
yticklabels(services);

```

OUTPUT:



MAIN PARAMETERS:

Available Spectrum → Divide into Bands → Assign Priority → Check Interference → Monitor Traffic → Dynamic Reallocation

Service	Frequency Range
IoT	700 – 900 MHz
LTE/4G	1800 – 2200 MHz
5G	3500 – 3800 MHz
Wi-Fi	5000 – 5200 MHz
Satellite	5800 – 6000 MHz

Conclusion:

Efficient frequency allocation reduces interference, improves spectrum utilization, and provides reliable communication for multiple wireless services.